

$$\dot{I}_3 \cdot (R_3 + jX_{L3})$$

$$= (4 - 2j) \cdot (5 e^{j\omega t})$$

$$\text{法二} \\ V_{ab} = \dot{I}_1 R_1 + L_1 \frac{d\dot{i}_1}{dt} + M \frac{d\dot{i}_2}{dt}$$

$$= \dot{I}_1 R_1 + L_1 j\omega \cdot \dot{I}_1 + M j\omega \cdot \dot{I}_2$$

$$= \dot{I}_1 R_1 + jX_{L1} \dot{I}_1 + jX_M \dot{I}_2$$

$$= (R_1 + jX_{L1}) \dot{I}_1 + jX_M \dot{I}_2$$

$$= (8 + 10j) \dot{I}_1 + 4j \cdot (-1 - 2j)$$

$$= 72 - 6j$$

法三,

$$V_{ab} = \dot{I}_1 \cdot [R_1 + j(X_{L1} + M)] + \dot{I}_3 \cdot (-jX_M)$$

$$= \cancel{4j} (3 - 4j)$$

$$= (3 - 4j)(8 + 14j) + (4 - 2j) \cdot (-4j)$$

$$= 72 - 6j$$